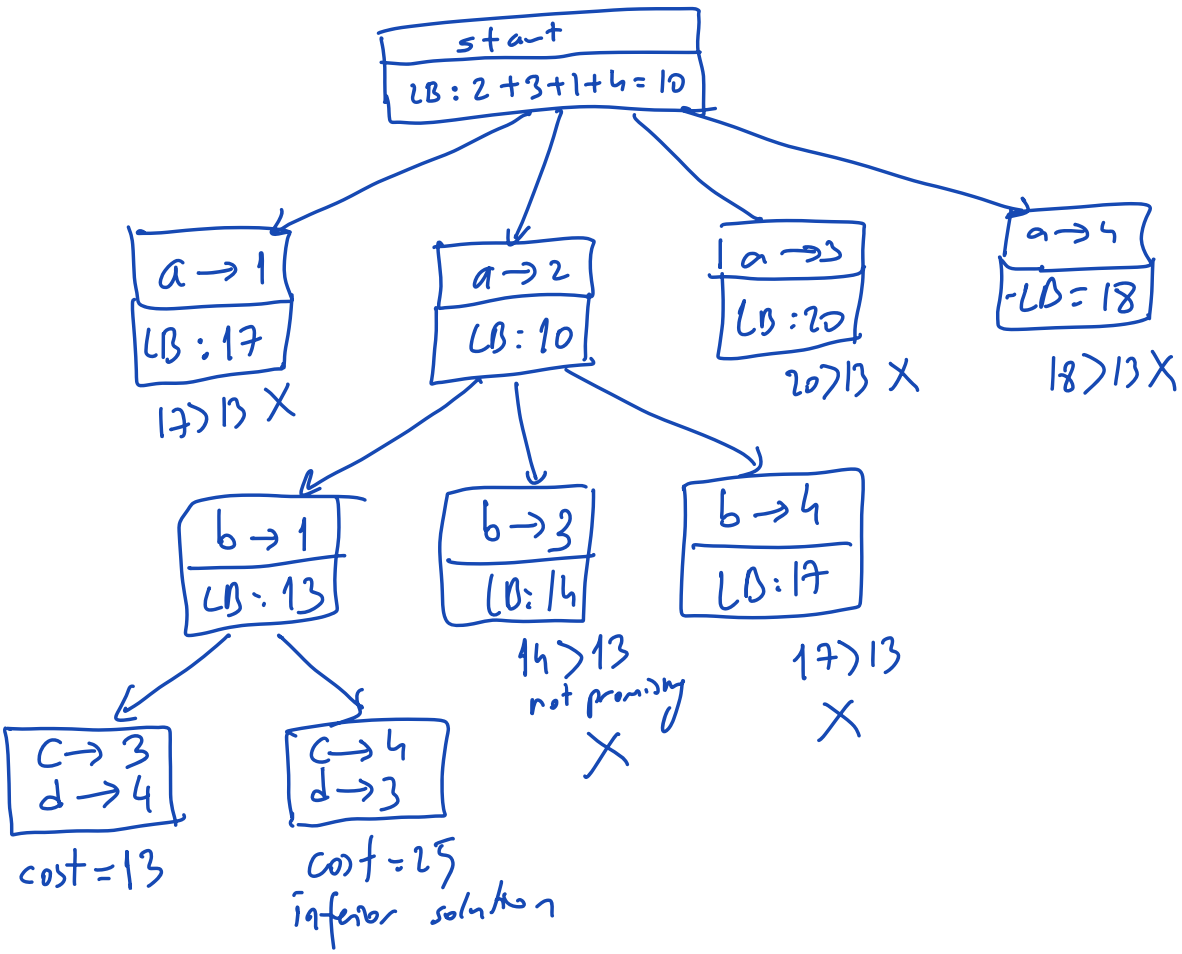
	Job 1	Job 2	Job 3	Job 4
Person *a*	9	2	7	8
Person *b*	6	4	3	7
Person *c*	5	8	1	8
Person *d*	7	6	9	4



	Job 1	Job 2	Job 3	Job 4
Person *a*	(9)	~~2~~	~~7~~	~~8~~
Person *b*	~~6~~	4	(3)	7
Person *c*	~~5~~	8	(1)	8
Person *d*	~~7~~	6	9	(4)

$a \rightarrow 1$

$lb = 9+3+1+4 = 17$

	Job 1	Job 2	Job 3	Job 4
Person *a*	~~9~~	2	~~7~~	~~8~~
Person *b*	6	~~4~~	3	7
Person *c*	5	~~8~~	1	8
Person *d*	7	~~6~~	9	4

$a \to 2$

$LB = 2+3+1+4 = 10$

	Job 1	Job 2	Job 3	Job 4
Person *a*	~~9~~	~~2~~	7	~~8~~
Person *b*	6	4	~~3~~	7
Person *c*	5	~~8~~	~~1~~	8
Person *d*	7	6	~~9~~	4

$a \to 3$

$LB: 7+4+5+4 = 20$

	Job 1	Job 2	Job 3	Job 4
Person *a*	~~9~~	~~2~~	7	~~8~~
Person *b*	6	4	~~3~~	7
Person *c*	5	~~8~~	~~1~~	8
Person *d*	7	~~6~~	~~9~~	4

$a \to 4$

$LB: 8+3+1+6 = 18$

	Job 1	Job 2	Job 3	Job 4
Person *a*	~~9~~	2	~~7~~	~~8~~
Person *b*	6	~~4~~	~~3~~	~~7~~
Person *c*	5	~~8~~	1	8
Person *d*	7	~~6~~	9	4

$a \to 2, b \to 1$

$LB: 2+6+1+4 = 13$

	Job 1	Job 2	Job 3	Job 4
Person *a*	~~9~~	2	~~7~~	~~8~~
Person *b*	~~6~~	~~4~~	3	~~7~~
Person *c*	5	~~8~~	~~1~~	8
Person *d*	7	~~6~~	9	4

$a \to 2 \quad b \to 3$

$LB: 2+3+5+4 = 14$

	Job 1	Job 2	Job 3	Job 4
Person *a*	~~9~~	2	~~7~~	~~8~~
Person *b*	~~6~~	~~4~~	~~3~~	7
Person *c*	~~5~~	~~8~~	1	~~8~~
Person *d*	7	~~6~~	~~9~~	4

$a \to 2 \quad b \to 4$

$LB: 2+7+1+7 = 17$

	Job 1	Job 2	Job 3	Job 4
Person *a*	9	2	7	8
Person *b*	6	4	3	7
Person *c*	5	8	1	8
Person *d*	7	6	9	4

a → 2 b → 1

c → 3 d → 4

cost = 13

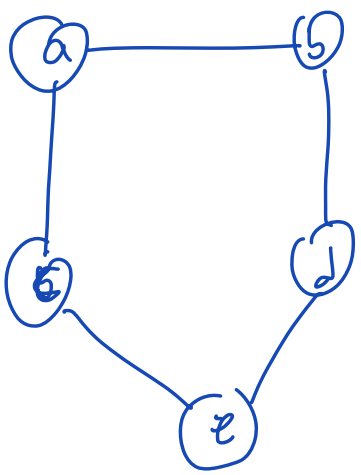
	Job 1	Job 2	Job 3	Job 4
Person *a*	9	2	7	8
Person *b*	6	4	3	7
Person *c*	5	8	1	8
Person *d*	7	6	9	4

a → 2 b → 1

c → 4 d → 3

cost = 2 + 6 + 8 + 9 = 25

Traveling salesman Problem

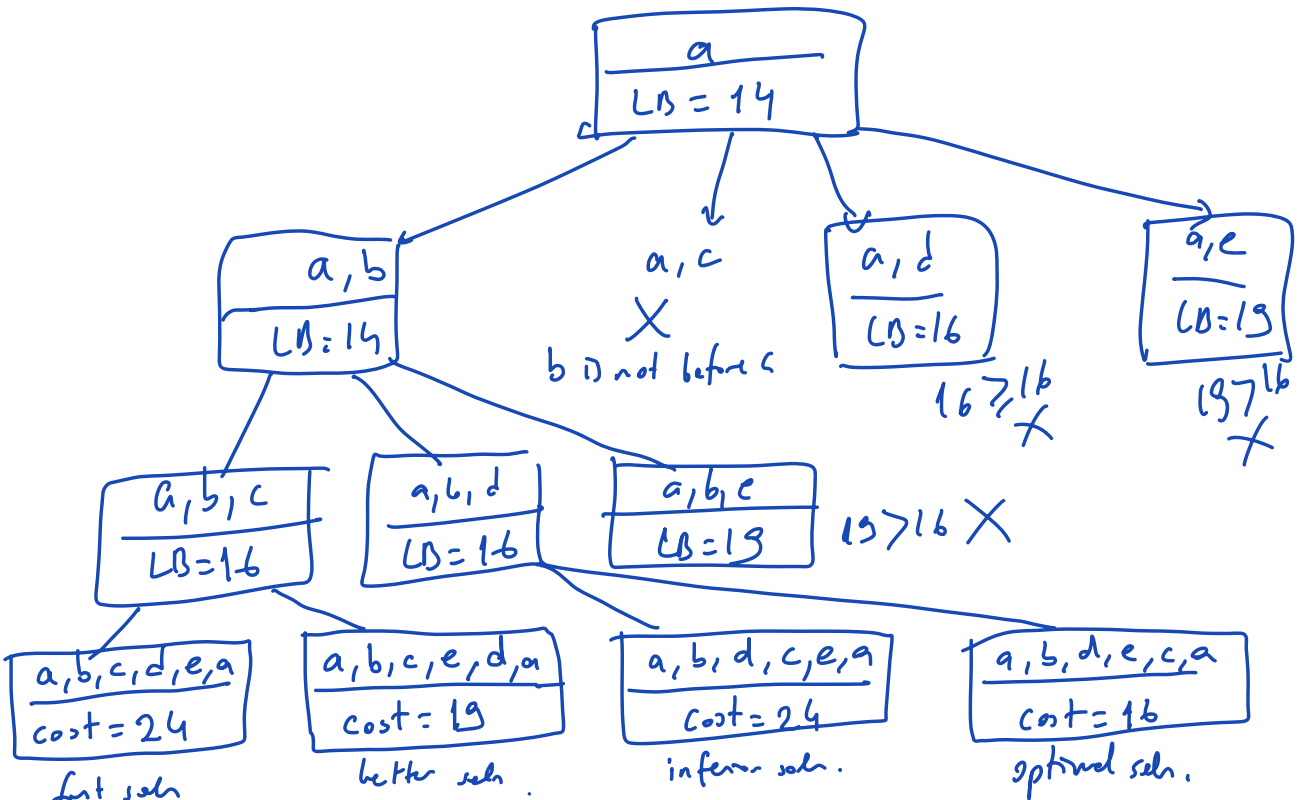


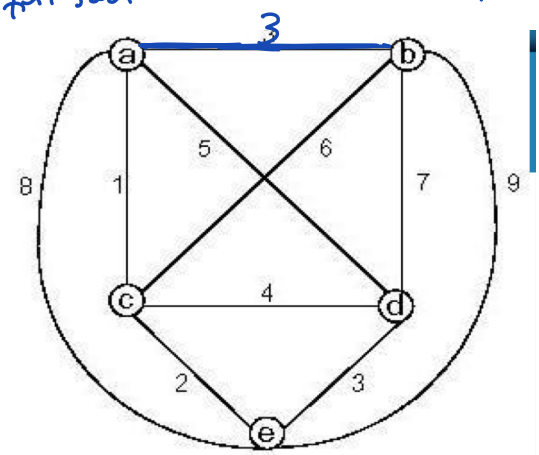
degree of any node = 2

$$\frac{\text{add all the two edges adjacent to each node}}{2}$$

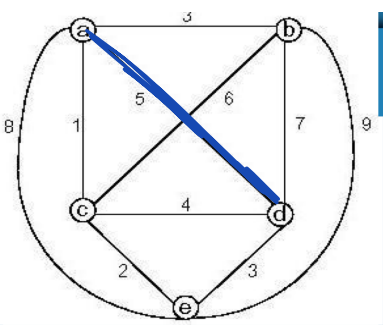
= tour length.

$$\frac{|ac| + |ab| + |ab| + |bd| + |bd| + |de| + |de| + |ce| + |ce| + |ac|}{2}$$



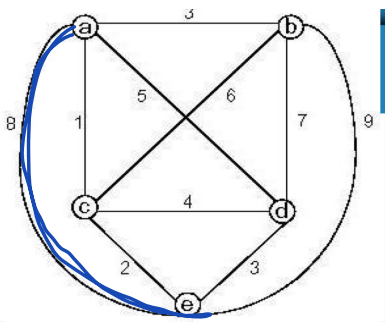


$$\left\lceil \frac{\overbrace{3+1}^{a} + \overbrace{3+6}^{b} + \overbrace{1+2}^{c} + \overbrace{3+5}^{d} + \overbrace{2+3}^{e}}{2} \right\rceil = 14$$

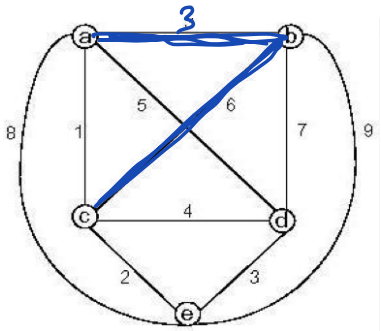


$$\left\lceil \frac{\overbrace{5+1}^{a} + \overbrace{3+6}^{b} + \overbrace{1+2}^{c} + \overbrace{5+3}^{d} + \overbrace{2+3}^{e}}{2} \right\rceil = 16$$

a, e



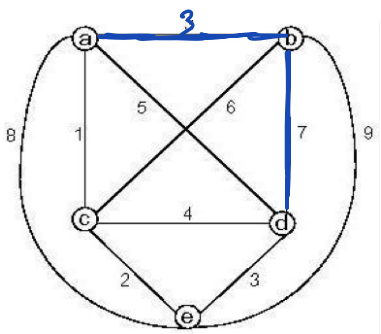
$$\left\lceil \frac{\overbrace{8+1}^{a} + \overbrace{3+6}^{b} + \overbrace{1+2}^{c} + \overbrace{3+4}^{d} + \overbrace{8+2}^{e}}{2} \right\rceil = 19$$



a, b, c

$$LB = \left\lceil \frac{\overbrace{3+1}^{a} + \overbrace{3+6}^{b} + \overbrace{6+1}^{c} + \overbrace{3+4}^{d} + \overbrace{2+3}^{e}}{2} \right\rceil$$

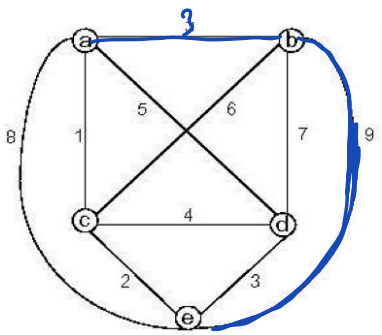
$$= 16$$



a, b, d

$$LB = \left\lceil \frac{\overbrace{3+1}^{a} + \overbrace{3+7}^{b} + \overbrace{1+2}^{c} + \overbrace{7+3}^{d} + \overbrace{2+3}^{e}}{2} \right\rceil$$

$$= 16$$



a, b, e

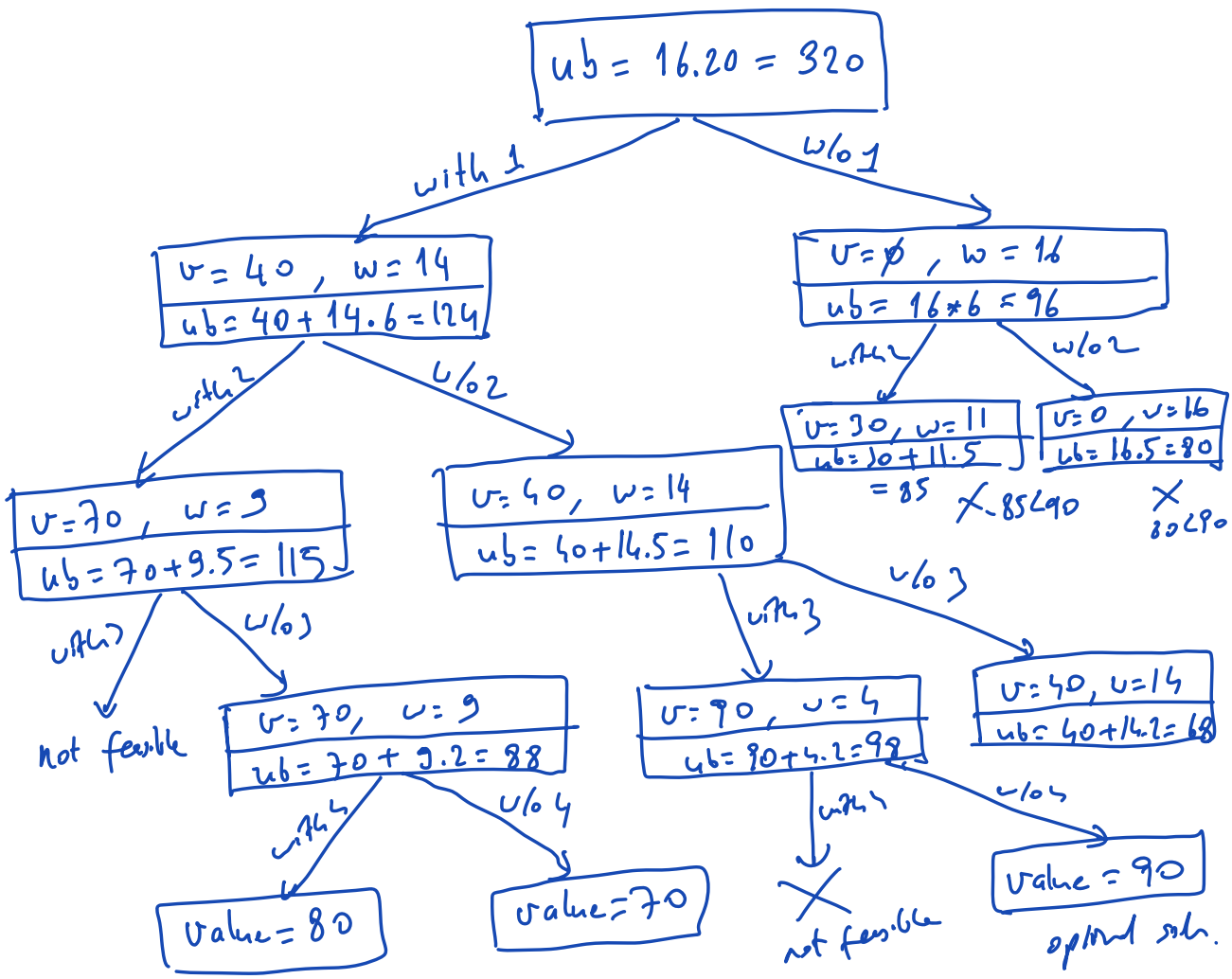
$$LB = \left\lceil \frac{\overbrace{3+1}^{a} + \overbrace{3+9}^{b} + \overbrace{1+2}^{c} + \overbrace{3+4}^{d} + \overbrace{9+2}^{e}}{2} \right\rceil$$

$$= 19$$

Knapsack Problem: (Branch & Bound approach)

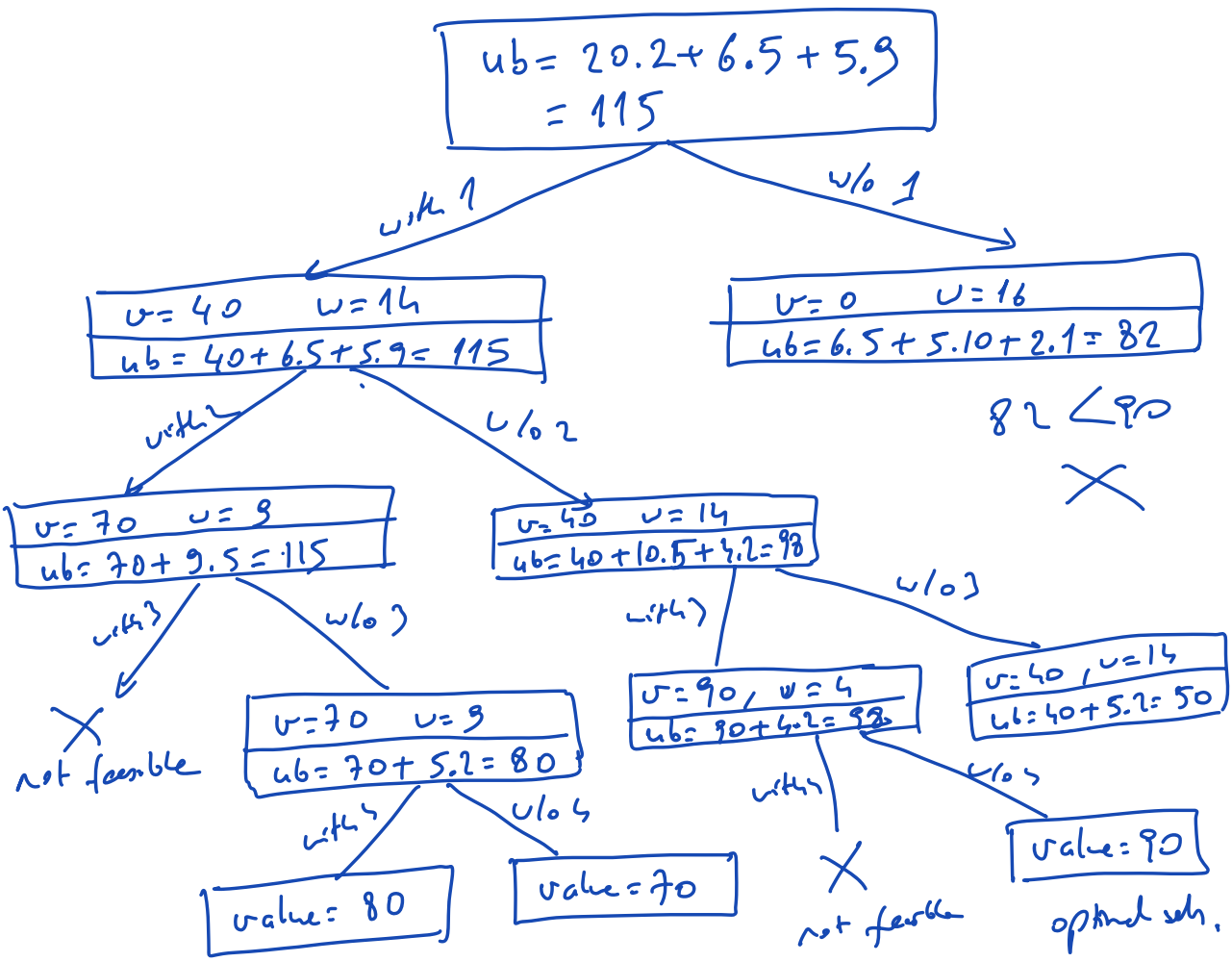
Example: The knapsack's capacity is 16

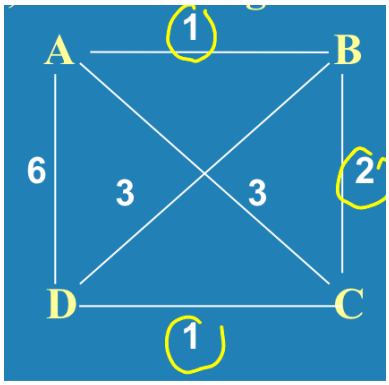
item	weight	value	v/w
1	2	$40	20
2	5	$30	6
3	10	$50	5
4	5	$10	2

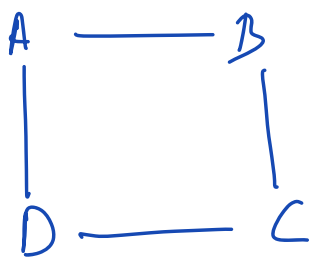


Let's consider another UB function. We say limit inclusion of each item by it's weight.

Example: The knapsack's capacity is 16			
item	weight	value	v/w
1	2	$40	20
2	5	$30	6
3	10	$50	5
4	5	$10	2







Multi-fragment Heuristic

(A, D) 1 ✓

(D, C) 1 ✓

(B, C) 2 ✓

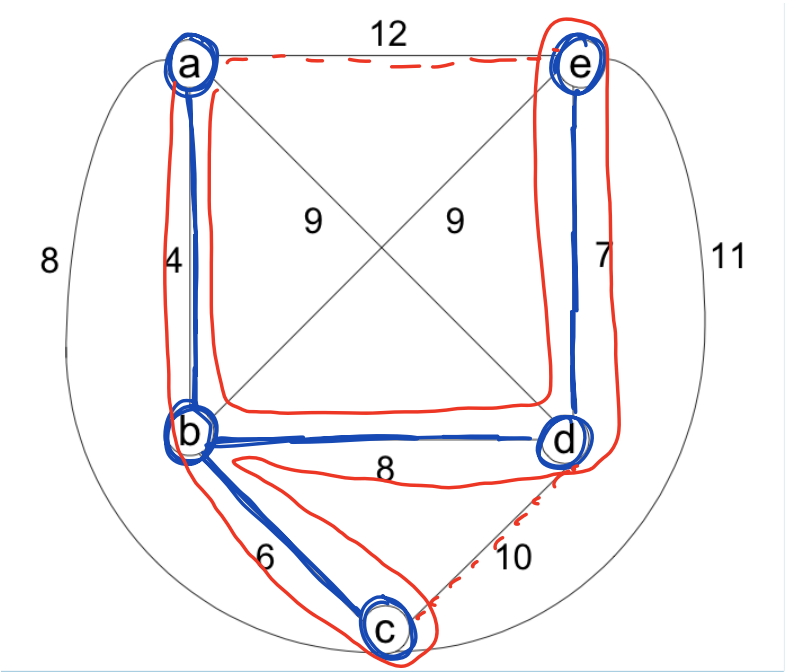
(A, C) 3 ✗ C's degree = 3

(B, D) 3 ✗ B's degree = 3

(A, D) 6 ✓

Length = 1 + 2 + 1 + 6 = 10

accuracy ratio = $\frac{10}{8}$ = 1,25

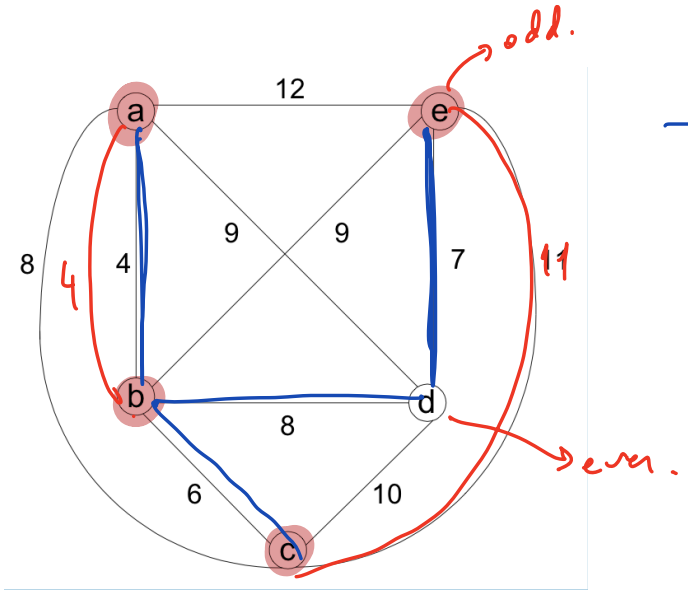


a b c b d e d b a ⇒ a b c d e a

For this instance, accuracy ratio = $\frac{39}{\text{optimal}}$.

This algorithm's performance ratio is ALSO unbounded above!

12 → arbitrary large. $\frac{\infty}{\text{optimal}}$



Christofides alg.

a-b = 4 b-c = 6 c-e = 11
a-c = 8 b-e = 9
a-e = 12

a-b, c-e : 4+11 = 15 → min.
a-c, b-e : 8+9 = 17
a-e, b-c : 12+6 = 18

a b c e d ~~a~~ a ⇒ Eulerian circuit.

a b c e d a ⇒ tour found by the alg.

total length: 37.

$W = 10$

item	w	v	v/w
1	1	20	20
2	10	190	19

optimal = 190 soln found = 20

accuracy ratio = 19/2

Example: The knapsack's capacity is 16			
item	weight	value	v/w
1	2	$40	20
2	5	$30	6
3	10	$50	5
4	5	$10	2

$k = 2$

$\phi \rightarrow 1, 2, 4 \rightarrow \$80$

$\{1\} \rightarrow 1, 2, 4 \rightarrow \$80$

$\{2\} \rightarrow 2, 1, 4 \rightarrow \$80$

$\{3\} \rightarrow 3, 1 \rightarrow \$90 \rightarrow$ best

$\{4\} \rightarrow 4, 1, 2 \rightarrow \$80$

$\{1,2\} \rightarrow 1, 2, 4 \rightarrow \$80$

$\{1,3\} \rightarrow 1, 3 \rightarrow \$90 \rightarrow$ best

$\{1,4\} \rightarrow 1, 4, 2 \rightarrow \$80$

$\{2,3\} \rightarrow 2, 3 \rightarrow \$80$

$\{2,4\} \rightarrow 2, 4, 1 \rightarrow \$80$

$\{3,4\} \rightarrow 3, 4 \rightarrow \$60$

Bin-Packing Problem



We have bins of capacity 1.
We have n items, each with weight $\in (0,1]$.
Goal is to place these items in min number of bins.

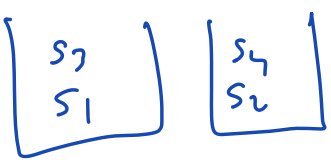
Example: $n = 4$, $s_1 = 0.4$ $s_2 = 0.2$ $s_3 = 0.6$ $s_4 = 0.7$

First-Fit alg:



3 bins

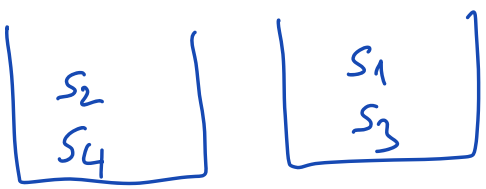
optimal:



2 bins accuracy ratio: $\frac{3}{2}$

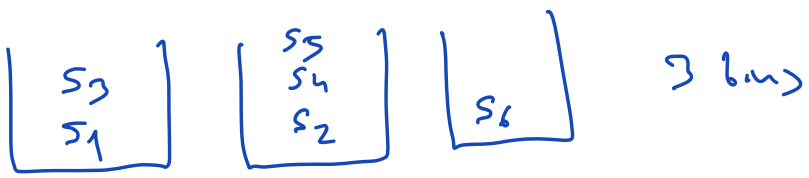
First-Fit Decreasing alg:

$s_4 = 0.7$ $s_3 = 0.6$ $s_1 = 0.4$ $s_2 = 0.2$



2 bins accuracy ratio: 1

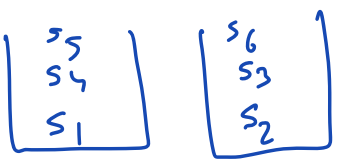
$s_1 = 0,6$ $s_2 = 0,5$ $s_3 = 0,3$ $s_4 = 0,2$ $s_5 = 0,2$ $s_6 = 0,2$



3 bins

accuracy ratio = $\frac{3}{2}$

optimal:



2 bins